

Spatial AI — The Stimuli Conjecture

A Minimal-Energy Perturbation Framework for Reflex-Dominant Cognitive Disruption

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Abstract

This paper presents a compile-safe research formulation of *The Stimuli Conjecture* for Spatial AI: within a sensorized environment, a sufficiently small and precisely timed stimulus can induce reflex-dominant motor action before conscious deliberation, producing predictable trajectory deviation without direct physical force. We model the environment as a spatiotemporal stimulus field, the human as a dual-timescale perceptual–motor system, and the interaction as a constrained stochastic control problem. We define reflex dominance, misstep magnitude, and a minimal-energy perturbation principle, and we provide a practical Phase-0 evaluation framework.

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1 Introduction

1.1 Doctrine Statement

Spatial AI Doctrine (Stimuli Form): The objective is not destruction. It is disruption of cognition. A sufficiently small, well-timed stimulus introduced into a space can compel reflex before reason. The environment delivers the stimulus; human instinct executes the misstep.

This document translates that doctrine into a mathematical framework.

1.2 Motivation

Human action in physical environments is shaped by layered response systems with different latencies:

- **Fast reflexive pathways** (subcortical / brainstem mediated),
- **Slower deliberative pathways** (cortical, attention-dependent, context-aware).

This temporal asymmetry creates an exploitable interval in which motion may initiate before conscious evaluation can intervene.

1.3 Research Goal

We seek a rigorous formulation of the problem:

Find the smallest (minimal-energy) spatiotemporal stimulus that causes a measurable, reliable trajectory deviation by producing reflex-dominant response prior to cognitive override.

2 Notation and System Model

2.1 Environment and Time

Let $\Omega \subset \mathbb{R}^3$ denote a bounded operational space and let $t \in [0, T]$ denote time.

Definition 2.1 (Stimulus Field). *A stimulus field is a measurable function*

$$S : \Omega \times [0, T] \rightarrow \mathbb{R}^k,$$

where each component corresponds to a stimulus modality (e.g., acoustic, photic, thermal, vibrotactile).

2.2 Human State

We model human state as

$$x_H(t) = [p(t), v(t), \theta(t), a(t), \sigma_c(t), \sigma_s(t)] \in \mathcal{X},$$

where:

- $p(t) \in \Omega$ is position,
- $v(t) \in \mathbb{R}^3$ is velocity,
- $\theta(t)$ is orientation / pose,
- $a(t) \in \mathbb{R}^3$ is acceleration,
- $\sigma_c(t) \in \mathbb{R}$ is cognitive load,
- $\sigma_s(t) \in \mathbb{R}$ is sensory susceptibility / arousal state.

2.3 Response Operator

Let $u(t) \in \mathbb{R}^m$ denote motor output. We write the response as

$$u(t) = \mathcal{R}(S(\cdot, \cdot), x_H(\cdot), \xi(t)),$$

where $\xi(t)$ captures latent variability (noise, internal state, surprise, stress, etc.).

Remark 2.2. *The operator $\mathcal{R}$ is generally unknown and should be estimated empirically. The framework remains valid for mechanistic, learned, or hybrid models.*

3 Dual-Timescale Perceptual–Motor Dynamics

3.1 Latency Separation

Let:

$$\tau_f > 0 \quad (\text{reflex latency}), \quad \tau_c > 0 \quad (\text{cognitive latency}),$$

with the key condition

$$\tau_f < \tau_c.$$

3.2 Two-Pathway Decomposition

We decompose motor output into fast and slow components:

$$u(t) = u_f(t) + u_c(t),$$

where u_f is reflexive and u_c is deliberative.

An abstract state-space form is:

$$\dot{z}_f(t) = A_f z_f(t) + B_f \phi_f(S_t, x_H(t)) + \eta_f(t), \quad (1)$$

$$\dot{z}_c(t) = A_c z_c(t) + B_c \phi_c(S_t, x_H(t)) + \eta_c(t), \quad (2)$$

$$u_f(t) = C_f z_f(t), \quad (3)$$

$$u_c(t) = C_c z_c(t), \quad (4)$$

where $S_t := S(\cdot, t)$.

3.3 Reflex Precedence Window

Define the reflex precedence window:

$$\mathcal{W}_{rp} := [\tau_f, \tau_c).$$

During $\mathcal{W}_{rp}$, motion may already be committed while cognitive verification is still incomplete.

4 Reflex Dominance and Disruption

4.1 Event Probabilities

For a stimulus policy S , define:

$$P_f(S \mid x_H) := \mathbb{P}(\text{motor initiation occurs within } \tau_f \mid x_H), \quad (5)$$

$$P_c(S \mid x_H) := \mathbb{P}(\text{cognitive override suppresses or corrects action before stabilization} \mid x_H). \quad (6)$$

Definition 4.1 (Reflex Dominance Ratio). *The reflex dominance ratio is*

$$D(S \mid x_H) := \frac{P_f(S \mid x_H)}{P_c(S \mid x_H) + \delta},$$

where $\delta > 0$ is a small regularization constant.

Definition 4.2 (Disruption Regime). *A stimulus is in the disruption regime if*

$$D(S \mid x_H) > 1.$$

It is reliably disruptive if

$$D(S \mid x_H) \geq \gamma,$$

for some reliability threshold $\gamma > 1$.

4.2 Misstep Magnitude

Let $T_0(t)$ be the nominal trajectory without stimulus and $T_S(t)$ the realized trajectory under S .

Definition 4.3 (Misstep Magnitude). *The misstep magnitude is*

$$M(S) := \int_0^T \|T_S(t) - T_0(t)\| dt.$$

Definition 4.4 (Cognitive Disruption Event). *A cognitive disruption event occurs when*

$$M(S) \geq \varepsilon,$$

where $\varepsilon > 0$ is a task-defined threshold.

5 Minimal-Energy Perturbation Principle

5.1 Stimulus Energy

We measure total stimulus energy by

$$\mathcal{E}(S) := \int_0^T \int_{\Omega} \|S(x, t)\|^2 dx dt.$$

5.2 Optimization Problem

Definition 5.1 (Minimal-Energy Disruptive Stimulus). *A minimal-energy disruptive stimulus is any optimizer*

$$S^* \in \arg \min_{S \in \mathcal{S}} \mathcal{E}(S)$$

subject to

$$M(S) \geq \varepsilon, \tag{7}$$

$$D(S \mid x_H) \geq \gamma, \tag{8}$$

$$S \in \mathcal{C}_{\text{safety}}, \tag{9}$$

$$S \in \mathcal{C}_{\text{hardware}}, \tag{10}$$

where $\mathcal{S}$ is the admissible stimulus space.

5.3 Penalty Relaxation

A soft-constrained version is:

$$\min_{S \in \mathcal{S}} \left[\mathcal{E}(S) + \lambda_1 \max(0, \varepsilon - M(S))^2 + \lambda_2 \max(0, \gamma - D(S \mid x_H))^2 + \lambda_3 \Xi_{\text{safety}}(S) \right].$$

6 The Stimuli Conjecture

Conjecture 6.1 (Spatial AI — The Stimuli Conjecture). *Let $\Omega \subset \mathbb{R}^3$ be a bounded operational space, and let the human perceptual-motor response exhibit dual-timescale dynamics with $\tau_f < \tau_c$. Then, for a broad class of operational states $x_H(t)$ and admissible actuator configurations, there exists a nontrivial stimulus field $S \not\equiv 0$ such that*

$$D(S \mid x_H) \geq \gamma, \tag{11}$$

$$M(S) \geq \varepsilon, \tag{12}$$

$$\mathcal{E}(S) \leq \rho, \tag{13}$$

for prescribed thresholds $\gamma > 1$, $\varepsilon > 0$, and sufficiently small energy budget $\rho > 0$.

6.1 Interpretation

The conjecture is an existence claim, not a universality claim. It states that a practical regime exists in which:

- timing precision compensates for low energy,
- context prediction compensates for stimulus intensity,
- environmental geometry amplifies perceived salience.

7 Assumptions and Identifiability

Assumption 7.1 (Latency Separation). *There exists a measurable latency gap:*

$$\tau_f + \Delta_{\text{jitter}} < \tau_c,$$

where Δ_{jitter} bounds timing uncertainty.

Assumption 7.2 (State Estimation). *The sensing system produces an estimate $\hat{x}_H(t)$ with bounded error:*

$$\|\hat{x}_H(t) - x_H(t)\| \leq \varepsilon_x$$

with high probability.

Assumption 7.3 (Actuation Feasibility). *The environment contains actuators capable of realizing $S \in \mathcal{S}$ within hardware and safety constraints.*

Assumption 7.4 (Nominal Trajectory Model). *A nominal trajectory $T_0(t)$ can be estimated for the target task class in no-stimulus conditions.*

7.1 Identifiability

The full response operator $\mathcal{R}$ need not be identified. Practical control requires predictive surrogates for:

1. $P_f(S \mid x_H)$,
2. $P_c(S \mid x_H)$,
3. $\mathbb{E}[M(S) \mid x_H]$.

8 Stochastic Control Formulation

8.1 Joint State Dynamics

Let the augmented state be

$$X(t) := (x_H(t), z_f(t), z_c(t)).$$

A generic controlled stochastic system can be written as

$$dX_t = f(X_t, S_t) dt + G(X_t) dW_t,$$

where W_t is a Wiener process and $S_t := S(\cdot, t)$.

8.2 Finite-Horizon Cost

Define:

$$J(S) = \mathbb{E} \left[\int_0^T \left(\alpha \|S_t\|_{L^2(\Omega)}^2 + \beta \ell_M(X_t) + \chi \ell_{\text{safety}}(S_t) \right) dt + \varphi(X_T) \right].$$

A constrained reliability form is:

$$\min_S \mathbb{E} \left[\int_0^T \|S_t\|_{L^2(\Omega)}^2 dt \right] \quad \text{subject to} \quad \mathbb{P}(M(S) \geq \varepsilon) \geq \eta, \quad \mathbb{E}[D(S \mid x_H)] \geq \gamma.$$

9 Spectral and Graph Formulation

9.1 Environmental Graph

Let the sensor-actuator topology be modeled as a weighted graph

$$G = (V, E), \quad |V| = n,$$

with adjacency W , degree matrix D_g , and Laplacian

$$L = D_g - W.$$

9.2 Eigenmode Stimulus Expansion

Let (λ_i, ϕ_i) be Laplacian eigenpairs:

$$L\phi_i = \lambda_i\phi_i.$$

A discrete stimulus can be expanded as

$$s(t) = \sum_{i=1}^n \alpha_i(t) \phi_i.$$

For continuous spaces, we may write

$$S(x, t) = \sum_{i=1}^N \alpha_i(t) \varphi_i(x),$$

where $\{\varphi_i\}$ are geometry-aligned basis functions.

Proposition 9.1 (Finite-Dimensional Reduction). *If the admissible stimulus space is truncated to*

$$S(x, t) = \sum_{i=1}^N \alpha_i(t) \varphi_i(x),$$

then the infinite-dimensional optimization over S reduces to a finite-dimensional control problem over coefficient trajectories $\alpha_1(t), \dots, \alpha_N(t)$.

Proof. Substitute the basis expansion into the objective $\mathcal{E}(S)$ and all constraints. The resulting quantities depend only on the coefficient trajectories and the basis Gram matrix. Therefore the optimization reduces to coefficient space. $\square$

10 Empirical Estimation and Phase-0 Metrics

10.1 Trial-Level Indicators

Given N trials indexed by j , define indicators

$$Y_f^{(j)} = \mathbf{1}\{\text{motor initiation within } \tau_f\}, \quad Y_c^{(j)} = \mathbf{1}\{\text{successful cognitive override}\}.$$

Empirical estimates are:

$$\hat{P}_f = \frac{1}{N} \sum_{j=1}^N Y_f^{(j)}, \quad \hat{P}_c = \frac{1}{N} \sum_{j=1}^N Y_c^{(j)},$$

and

$$\hat{D} = \frac{\hat{P}_f}{\hat{P}_c + \delta}.$$

10.2 Primary Metrics

1. **Reflex activation probability:** $\hat{P}_f$
2. **Override probability:** $\hat{P}_c$
3. **Reflex dominance ratio:** $\hat{D}$
4. **Misstep magnitude:** $\hat{M}$
5. **Stimulus energy:** $\hat{\mathcal{E}}(S)$
6. **Latency shift:** $\Delta\hat{\tau}$ (relative to no-stimulus baseline)

10.3 Phase-0 Success Criterion

A Phase-0 demonstration is considered successful if, under controlled and repeatable conditions,

$$\hat{D} \geq \gamma \quad \text{and} \quad \hat{M} \geq \varepsilon$$

while maintaining

$$\hat{\mathcal{E}}(S)$$

as low as possible and within safety constraints.

11 Algorithmic Sketch for Stimulus Policy Search

11.1 Practical Loop

A practical closed-loop implementation can follow:

1. Sense and estimate human state $\hat{x}_H(t)$.
2. Predict reflex susceptibility and override likelihood for candidate stimuli.
3. Solve a short-horizon constrained optimization for S_t .
4. Apply the stimulus under hardware/safety limits.
5. Measure outcome and update surrogate models.

11.2 Pseudo-Objective

At each control step t , solve approximately:

$$S_t^* \in \arg \min_{S_t \in \mathcal{U}} \left[\alpha \|S_t\|_{L^2(\Omega)}^2 + \lambda_1 \max(0, \varepsilon - \widehat{M}(S_t, \hat{x}_H))^2 + \lambda_2 \max(0, \gamma - \widehat{D}(S_t, \hat{x}_H))^2 \right].$$

12 Scope, Limits, and Research Notes

12.1 What This Document Claims

This document claims:

- a formal problem statement,
- a measurable doctrine-to-math translation,
- an optimization framework for minimal-energy stimulus design,
- a testable Phase-0 evaluation structure.

12.2 What This Document Does Not Claim

This document does not claim:

- universal effectiveness across all populations or conditions,
- a validated physiological parameter set,
- an experimentally proven theorem (the central statement remains a conjecture).

13 Conclusion

Spatial AI can be framed as a control problem over perception, timing, and environment geometry. The central principle is reflex precedence: minimal, well-timed stimuli can produce measurable trajectory deviation before deliberative correction engages.

The Stimuli Conjecture formalizes this as a minimal-energy perturbation problem subject to reflex-dominance and misstep constraints. The result is a clean doctrine-to-math bridge suitable for Phase-0 research planning, simulation, and experimental design.

A Symbol Table

Symbol	Meaning
Ω	Bounded operational space in $\mathbb{R}^3$
T	Time horizon
$S(x, t)$	Stimulus field (spatiotemporal, multimodal)
$x_H(t)$	Human state (position, velocity, pose, load, susceptibility)
$u(t)$	Motor output
$\mathcal{R}$	Perceptual–motor response operator
τ_f	Reflex latency
τ_c	Cognitive latency
P_f	Probability of reflexive motor initiation
P_c	Probability of cognitive override
$D(S \mid x_H)$	Reflex dominance ratio
$T_0(t)$	Nominal no-stimulus trajectory
$T_S(t)$	Realized trajectory under stimulus
$M(S)$	Misstep magnitude / trajectory deviation functional
ε	Minimum disruption threshold
$\mathcal{E}(S)$	Stimulus energy functional
γ	Reliability threshold for disruption
ρ	Energy budget bound
L	Graph Laplacian for sensor-actuator topology
ϕ_i	Laplacian eigenvector / mode
$\alpha_i(t)$	Time-varying coefficient for mode i